

Solar Panel Spec Sheets

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BP 365

65 Watt Photovoltaic Module

High-efficiency photovoltaic module using silicon nitride multicrystalline silicon cells.

Performance

Rated power (P_{max})	65W
Nominal voltage	12V
Limited Warranty ¹	25 years

Configuration

J Clear universal frame and standard J-Box

Electrical Characteristics²

BP 365

Maximum power (P_{max}) ³	65W
Voltage at Pmax (V_{mp})	17.6V
Current at Pmax (I_{mp})	3.69A
Warranted minimum P_{max}	60W
Short-circuit current (I_{sc})	3.99A
Open-circuit voltage (V_{oc})	21.7V
Temperature coefficient of I_{sc}	(0.065±0.015)%/ °C
Temperature coefficient of V_{oc}	-(80±10)mV/°C
Temperature coefficient of power	-(0.5±0.05)%/ °C
NOCT (Air 20°C; Sun 0.8kW/m ² ; wind 1m/s)	47±2°C
Maximum series fuse rating	20A
Maximum system voltage	600V (ETL & IEC61215 rating)



Mechanical Characteristics

Dimensions	Length: 796mm (31.33") Width: 674mm (26.54") Depth: 50mm (1.97")
Weight	6.4 kg (14.11 pounds)
Solar Cells	36 cells (156mm x 78mm) in a 4x9 matrix connected in series
Junction Box	J-Version junction box with 4-terminal connection block; IP 65, accepts PG 13.5, M20, ½ inch conduit, or cable fittings accepting 6-12mm diameter cable. Terminals accept 2.5 to 10mm ² (8 to 14 AWG) wire.
Diodes	IntegraBus™ technology includes Schottky by-pass diodes integrated into the printed circuit board bus
Construction	Front: High-transmission 3mm (1/8 th inch) tempered glass; Back: White Polyester; Encapsulant: EVA
Frame	Clear anodized aluminum alloy type 6063T6 Universal frame; Color: silver

1. Module Warranty: 25-year limited warranty of 80% power output; 12-year limited warranty of 90% power output; 5-year limited warranty of materials and workmanship. See your local representative for full terms of these warranties.
2. This data represents the performance of typical BP modules, and are based on measurements made in accordance with ASTM E1036 corrected to SRC (STC.)
3. During the stabilization process that occurs during the first few months of deployment, module power may decrease by approx. 1% from typical P_{max} .

ELECTRICAL SPECIFICATIONS



Model Type	ET-M660260WW	ET-M660255WW	ET-M660250WW	ET-M660245WW	ET-M660240WW	ET-M660235WW
Peak Power (Pmax)	260W	255W	250W	245W	240W	235W
Module Efficiency	15.98%	15.67%	15.37%	15.06%	14.75%	14.44%
Maximum Power Voltage (Vmp)	31.14V	30.91V	30.43V	30.08V	30.12V	29.69V
Maximum Power Current (Imp)	8.35A	8.25A	8.22A	8.15A	8.02A	7.92A
Open Circuit Voltage (Voc)	37.86V	37.82V	37.70V	37.40V	37.37V	37.01V
Short Circuit Current (Isc)	8.96A	8.88A	8.69A	8.61A	8.60A	8.49A
Power Tolerance	-1% to +3%	0 to +5W	0 to +5W	0 to +5W	0 to +5W	0 to +5W
Maximum System Voltage	DC 1000V					
Nominal Operating Cell Temperature	44.4±2°C					
Series Fuse Rating (A)	20A					
Number of Bypass Diode	3					

MECHANICAL SPECIFICATIONS

Cell type	156mm x 156mm
Number of cells	60 cells in series
Weight	19.32kg(44.59lbs)

Dimensions 1640×992×40 mm
(64.57×39.06×1.57 inch)

Max Load	5400Pascals (112 lb/ft²)
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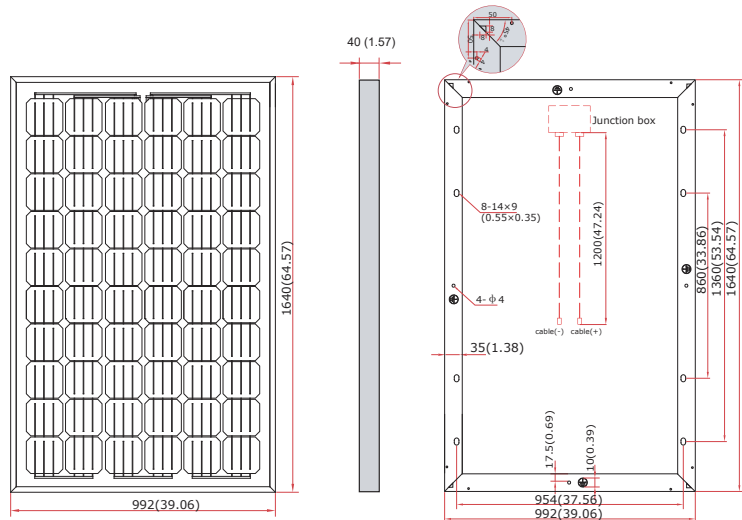
TEMPERATURE COEFFICIENT

Temp. Coeff. of Isc (TK Isc)	0.02 %/°C
Temp. Coeff. of Voc (TK Voc)	-0.31 %/°C
Temp. Coeff. of Pmax (TK Pmax)	-0.44 %/°C

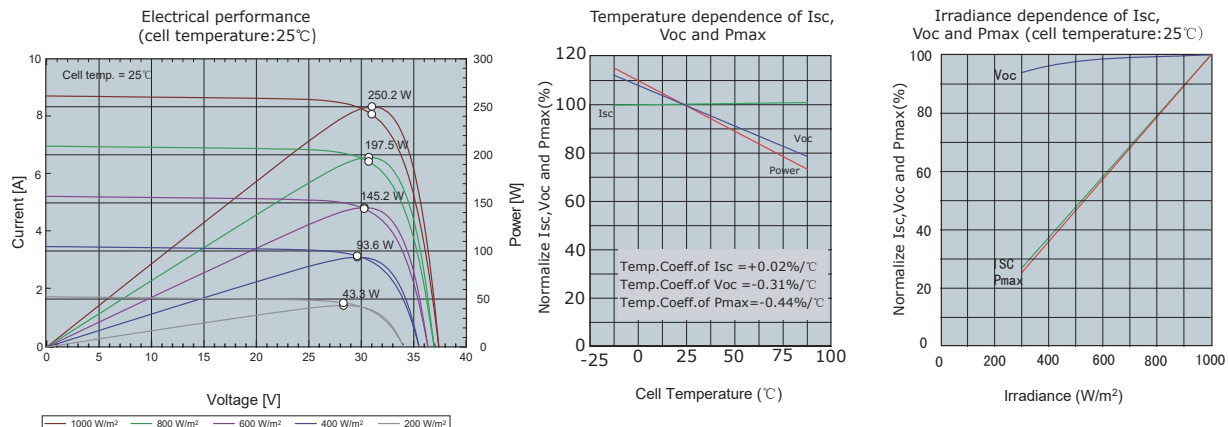
BIG CARTON PACKING MANNER

Container	20' GP	40' GP
Pieces per Pallet	26	26
Pieces per Container	312	728

PHYSICAL CHARACTERISTICS Unit:mm (inch)



ELECTRICAL CHARACTERISTICS



Note: the specifications are obtained under the Standard Test Conditions (STCs): 1000 W/m² solar irradiance, 1.5 Air Mass, and cell temperature of 25 °C. The NOCT is obtained under the Test Conditions : 800 W/m², 20°C ambient temperature, 1 m/s wind speed, AM 1.5 spectrum.

Please contact support@etsolar.com for technical support. The parameters are for reference only, and are subject to change without notice or obligation.

HSL60 | Poly UL

Electrical Characteristics

Electrical Characteristics at Standard Test Conditions (STC)

Power Class	235 W	240 W	245 W
Maximum Power (P_{max})	235 W	240 W	245 W
Open Circuit Voltage (V_{oc})	37.1V	37.3V	37.4V
Short Circuit Current (I_{sc})	8.51A	8.58A	8.63A
Voltage at Maximum Power (V_{mpp})	29.8V	30.0V	30.4V
Current at Maximum Power (I_{mpp})	7.89A	8.02A	8.08A
Module Efficiency (%)	14.12%	14.42 %	14.72%
Cell Efficiency (%)	16.59%	16.94 %	17.29%

P_{max} , V_{oc} , I_{sc} , V_{mpp} , and I_{mpp} tested at STC defined as irradiance of 1000 W/m² at AM 1.5 solar spectrum and temperature 25 ± 2 °C.
Electrical Characteristics: measurement tolerance of ± 3 %.

Electrical Characteristics at Normal Operating Cell Temperature (NOCT)

Power Class	235 W	240 W	245 W
Maximum Power (P_{max})	171W	175W	178W
Open Circuit Voltage (V_{oc})	34.1V	34.3V	34.4V
Short Circuit Current (I_{sc})	6.88A	6.94A	6.98A
Voltage at Maximum Power (V_{mpp})	27.0V	27.1V	27.5V
Current at Maximum Power (I_{mpp})	6.33A	6.43A	6.48A
Module Efficiency (%)	10.28%	10.52%	10.70%

P_{max} , V_{oc} , I_{sc} , V_{mpp} , and I_{mpp} tested at NOCT defined as irradiance of 800 W/m²; wind speed 1 m/s.
Electrical Characteristics: measurement tolerance of ± 3 %.

Temperature Characteristics

Temperature Coefficients of P	- 0.43 %/°C
Temperature Coefficients of V	- 0.31 %/°C
Temperature Coefficients of I	+ 0.056 %/°C

Maximum Ratings

Maximum System Voltage	600 V (UL & CUL)
Series Fuse Rating	15 A
Maximum Reverse Current	1.35×Fuse rating

Nomenclature

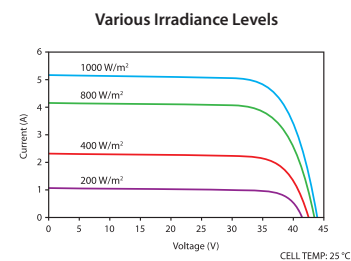
Full product name:

HSL60P-PA-0-xxxK(UL)

xxx represents the power class

Performance at Low Irradiance:

The typical relative change in module efficiency at an irradiance of 200 W/m² in relation to 1000 W/m² (both at 25 °C and AM 1.5G spectrum) is less than 5 %.



Mechanical Characteristics

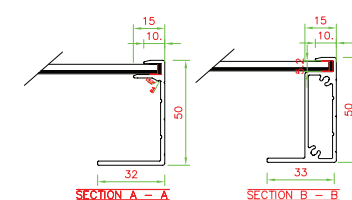
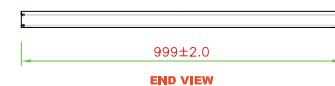
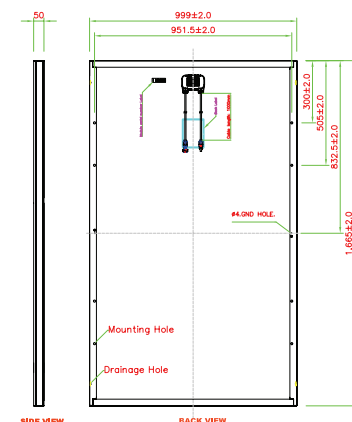
Dimensions	1665mm × 999mm × 50 mm
Weight	20kg
Frame	Anodized aluminum frame
Front	3.2mm low iron tempered glass
Encapsulant	EVA
Back Cover	White back sheet
Cell Technology	Poly-Si
Cell Size	156mm × 156 mm
Number of Cells (Pieces)	60 (6 × 10)
Junction Box	Protection class IP65, with bypass-diode
Output Cables	Solar cable: 4 mm ² ; length: 1000 mm
Connector	MC4 compatible

System Design

Operating Temperature	- 40 °C to 85 °C
Hail Safety Impact Velocity	Hailstone (25mm) at 23m/s
Fire Safety Classification (IEC 61730)	Class C
Static Load Wind / Snow	2400 Pa / 5400 Pa

Packaging and Storage

Storage Temperature	- 40 °C to 85 °C
Packaging Method	Sleeve packing

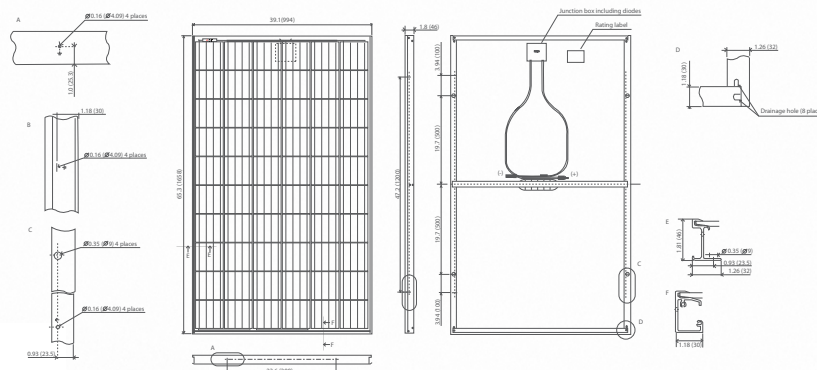


Mitsubishi Electric Photovoltaic Module

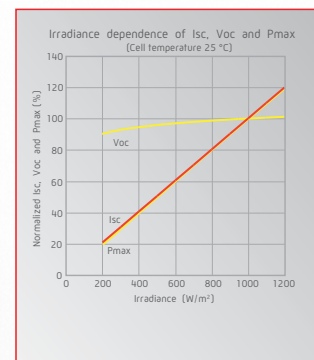
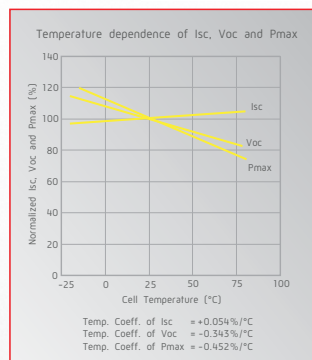
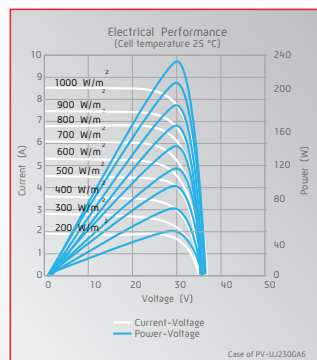
Specification Sheet

Manufacturer	MITSUBISHI ELECTRIC				
Model name	PV-UJ235GA6	PV-UJ230GA6	PV-UJ225GA6	PV-UJ218G6	PV-UJ212G6
Cell type	Polycrystalline Silicon, 156 mm x 156 mm				
Number of cells	60 cells in a series				
Maximum power rating (Pmax)	235Wp	230Wp	225Wp	218Wp	212Wp
Warranted minimum Pmax	228.0Wp	223.1Wp	218.3Wp	211.5Wp	205.6Wp
Tolerance of maximum power rating	+3/-3%	+3/-3%	+3/-3%	+3/-3%	+3/-3%
PV USA test condition rating (PTC)	TBD	TBD	TBD	TBD	TBD
Open circuit voltage (Voc)	36.8V	36.6V	36.4V	36.3V	36.1V
Short circuit current (Isc)	8.49A	8.39A	8.30A	8.25A	8.13A
Maximum power voltage (Vmp)	30.5V	30.2V	30.0V	29.3V	29.0V
Maximum power current (Imp)	7.71A	7.62A	7.50A	7.44A	7.31A
Static load test passed	5,400 Pa	5,400 Pa	5,400 Pa	2,400 Pa	2,400 Pa
Number of bus bar per cell	4 Bus bars	4 Bus bars	4 Bus bars	2 Bus bars	2 Bus bars
Module efficiency	14.3%	14.0%	13.7%	13.2%	12.9%
Normal operating cell temperature (NOCT)	47.5 degree C - Under evaluation				
Maximum system voltage	DC 600V				
Fuse rating	15A				
Dimensions	65.3 x 39.1 x 1.81 inch (1658 x 994 x 46 mm)				
Weight	44 lbs (20kg)				
Number of modules per pallet	20				
Number of modules per container (40 ft. container)	560				
Packing contents	2 pcs - 1 carton				
Output terminal	[Positive] 1025mm [Negative] 1025mm with MC connector (PV-KBT4/6 II-UR, PV-KST4/6 II-UR)				
Certifications	IEC 61215 2nd Edition, UL61730, conforms to UL1703				
Fire rating	Class C				

Drawings and Dimensions Unit: inch (mm)



Electrical Characteristics



Mitsubishi Electric & Electronics USA, Inc.
 5665 Plaza Drive, Cypress, CA 90630-0007
 Telephone: 714-220-2500 Email: pv@meus.mea.com
www.mitsubishielectricsolar.com

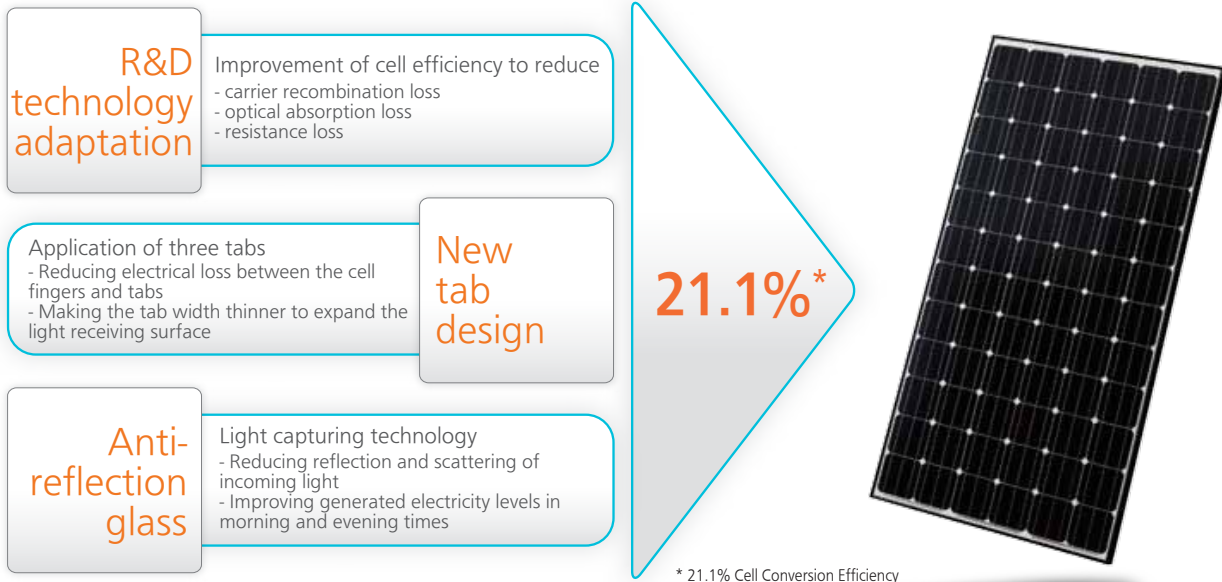
Panasonic

ideas for life

HIT Photovoltaic Module

HIT Power 235S

VBHN235SA06



HIT[®] Power

Photovoltaic Module

Ultra High Efficiency & Superior Real World Performance

HIT Power Solar panels are leaders in sunlight conversion efficiency. Our hybrid cells produce the highest output on cloudy days.

21.1% Cell Conversion Efficiency

Most PTC Power 218.7W

Highest PTC/STC Ratio 93% +

Superior Temperature Performance

As temperatures rise, HIT Power solar panels produce 10% or more electricity (kWh) than conventional crystalline silicon solar panels at the same temperature.

Quality and Reliability

Panasonic is truly committed to quality since it began developing and manufacturing solar PV in 1975. Since pioneering, developing and launching HIT Solar cells in the 1990s, we have been the technology leader, and for decades many satisfied customers have placed their trust in the competence in our unique HIT Technology.

Power Guarantee

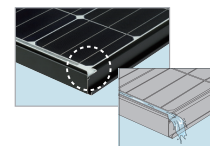
The power ratings for HIT Power panels guarantee customers receive 100% of the nameplate rated power (or more) at the time of purchase, enabling owners to generate more kWh per rated Watt, quicken investments returns, and help realize complete customer satisfaction.

American Made Quality

Our silicon wafers located inside HIT solar panels are made in Oregon, and the panels are assembled in an ISO 9001 (quality), 14001 (environment), and 18001 (safety) certified factory. Unique eco-packing minimizes cardboard waste at the job site. The panels have a Limited 20-Year Power Output and 10-Year Product Workmanship Warranty.

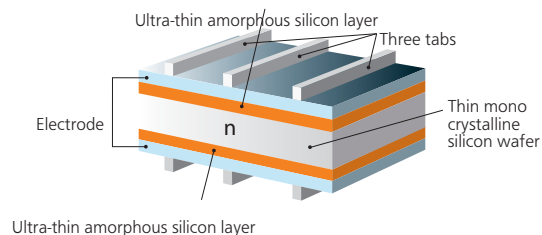
"Drainage corner frame"

to prevent the accumulation of dirt on the panel



Rainwater flush the dirt of the panel

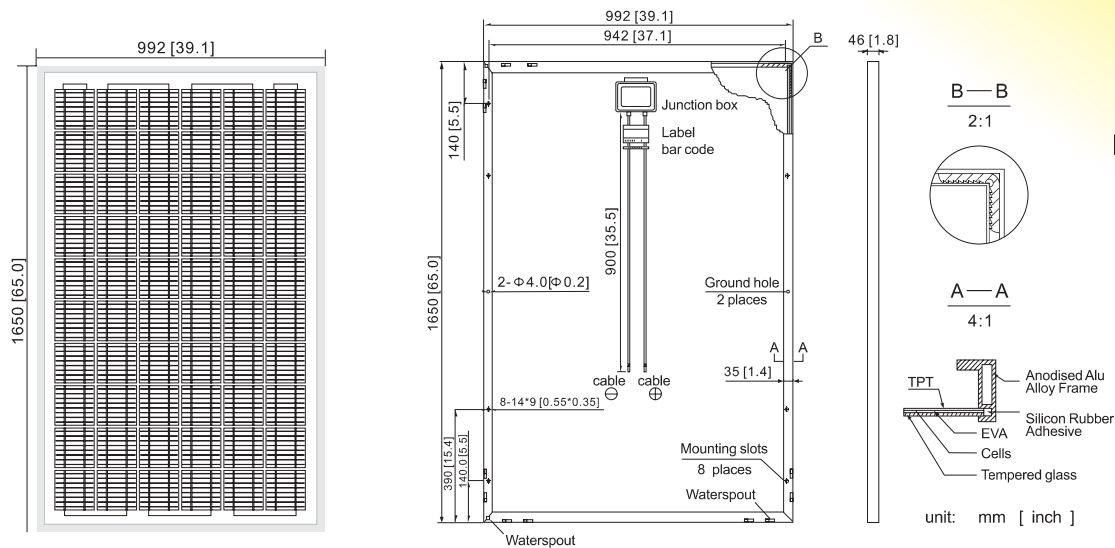
HIT[®] solar cell structure



HIT solar cells are hybrids of mono crystalline silicon surrounded by ultra thin amorphous silicon layers, and are available solely from Panasonic.

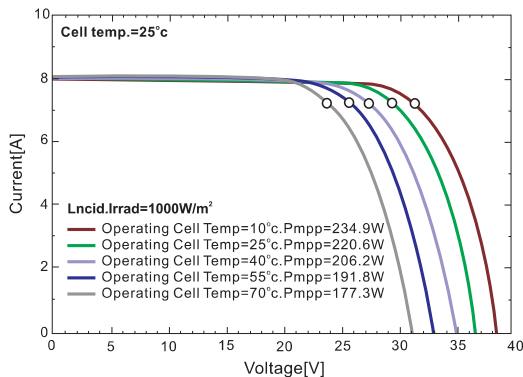
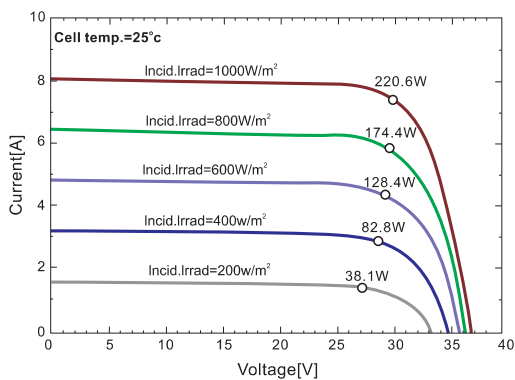
Perlight Solar PLM P-60

Module Diagram



PERLIGHT

IV Curves



Mechanical Characteristics

Cell Size (in)	6 x 6
Module Size (in)	65.0 x 39.1 x 1.8
# of cells	60 (6x10)
Weight	46 lbs : 21 kg
# of mounting holes	8
# of waterspouts	16

Temperature Coefficients

NOCT	47 ± 2°C
I _{sc}	+0.05%/°C
V _{oc}	-0.35%/°C
P _m	-0.45%/°C
Power Tolerance	+/- 3%

Electrical Characteristics

Model	PLM-225P-60	PLM-230P-60	PLM-235P-60	PLM-240P-60
P _m	225	230	235	240
V _{mp}	29.9	30.0	30.3	30.6
I _{mp}	7.53	7.66	7.75	7.84
V _{oc}	36.9	37.0	37.3	37.5
I _{sc}	8.09	8.18	8.28	8.38
Max sys voltage	1000v	1000V	1000V	1000V
Module Efficiency	13.7%	14.1%	14.4%	14.7%
PTC rating	197.4	206.4	210.9	214.6

*outputs given at STC 1,000 W/m²
25°C AM 1.5

PERLIGHT SOLAR

400 Continental Blvd, 6th Fl
El Segundo CA 90245
(424) 242 8068

info@PerlightUSA.com
www.PerlightUSA.com

This publication summarizes product warranty and specifications, which are subject to change without notice. Additional information is found on our website.

12230 Eastern Ave., Chase, Maryland 21220

Phone: 410-344-0800 Toll Free: 866-346-1391 Fax: 410-344-9240



Preliminary Product Information Sheet Solar Module BSP60/65-12

General Information

High performance multi-crystalline cells
High energy yields in a wide variety of climates
Anodized aluminum frame provides for easy mounting

Qualification and Certificates

UL Listing Pending

Electrical Characteristics

Typical Data At STC			
		BSP60-12	BSP65-12
Rated Power	Pr	60W	65W
Peak Power	Pmpp	60W	65W
Peak Power Voltage	Vmpp	16.9V	17.4V
Peak Power Current	Impp	3.55A	3.75A
Open Circuit Voltage	Voc	21.5V	21.7V
Short Circuit Current	Isc	3.73A	3.99A

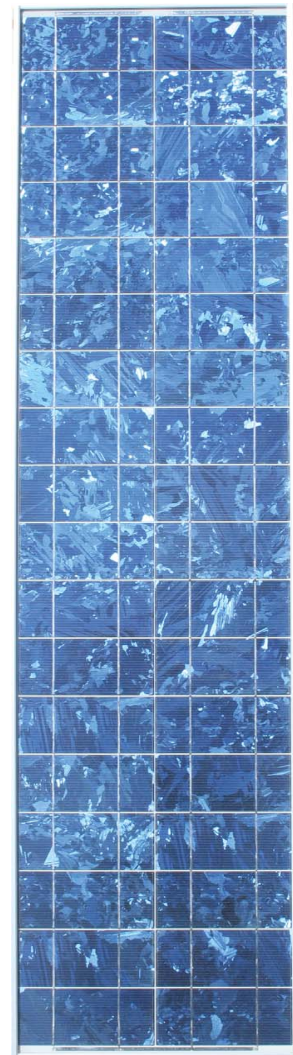
The performance of the solar cells is measured at Standard Test Conditions (STC): 1000 W/m² irradiance, AM 1.5 spectrum and 25° C cell temperature.

Physical Characteristics

Length	Inches (mm)	58.0 (1473.2)
Width	Inches (mm)	13.75 (349.25)
Thickness	Inches (mm)	1.75 (44.4)

Cable Type	18/2	
Cable Length	feet (cm)	15.0 (457.2)

10 Year Warranty



Sharp ND-240QCJ

240 WATT

ND-240QCJ

Module output cables: 12 AWG PV Wire (per UL Subject 4703)

ELECTRICAL CHARACTERISTICS

Maximum Power (Pmax)*	240 W
Tolerance of Pmax	+5%/-0%
PTC Rating	216.4 W
Type of Cell	Polycrystalline silicon
Cell Configuration	60 in series
Open Circuit Voltage (Voc)	37.5 V
Maximum Power Voltage (Vpm)	29.3 V
Short Circuit Current (Isc)	8.75 A
Maximum Power Current (Ipm)	8.19 A
Module Efficiency (%)	14.7%
Maximum System (DC) Voltage	600 V (UL)/1000V (IEC)
Series Fuse Rating	15 A
NOCT	47.5°C
Temperature Coefficient (Pmax)	-0.485%/°C
Temperature Coefficient (Voc)	-0.36%/°C
Temperature Coefficient (Isc)	0.053%/°C

*Illumination of 1 kW/m² (1 sun) at spectral distribution of AM 1.5 (ASTM E892 global spectral irradiance) at a cell temperature of 25°C.

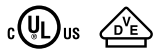
MECHANICAL CHARACTERISTICS

Dimensions (A x B x C to the right)	39.1" x 64.6" x 1.8"/994 x 1640 x 46 mm
Cable Length (G)	43.3"/1100 mm
Output Interconnect Cable	12 AWG with *SMK Locking Connector
Hail Impact Resistance	1" (25 mm) at 52 mph (23 m/s)
Weight	41.9 lbs / 19.0 kg
Max Load	50 psf (2400 Pascals)
Operating Temperature (cell)	-40 to 194°F / -40 to 90°C

*Intertek recognized for mating with MC-4 connectors (part numbers PV-KST4; PV-KBT4)

CERTIFICATIONS

UL 1703, ULC/ORD-C1703, IEC 61215, IEC 61730, CEC, FSEC



WARRANTY

25-year limited warranty on power output
Contact Sharp for complete warranty information

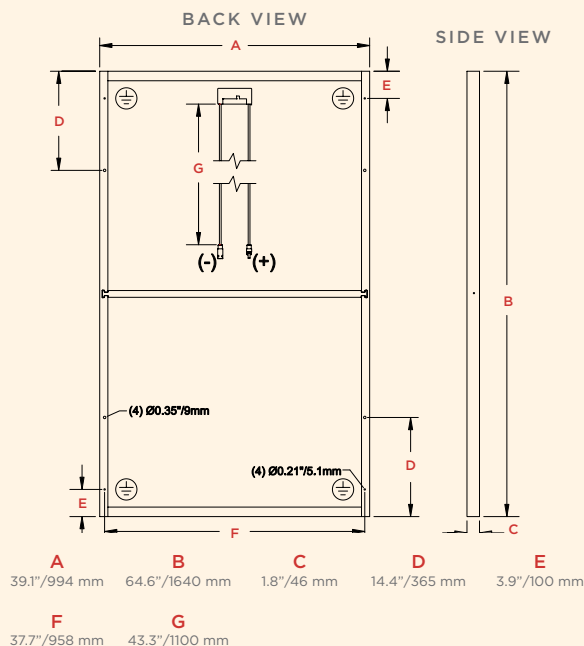
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SHARP®

SHARP ELECTRONICS CORPORATION
5700 NW Pacific Rim Boulevard, Camas, WA 98607
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www.sharpsolar.com

DIMENSIONS



Contact Sharp for tolerance specifications

ISO QUALITY & ENVIRONMENTAL MANAGEMENT

Sharp solar modules are manufactured in ISO 9001:2000 AND ISO 14001:2004 certified facilities.

"BUY AMERICAN"

Sharp solar modules are manufactured in the United States and Japan, and qualify as "American" goods under the "Buy American" clause of the American Recovery and Reinvestment Act (ARRA).



Solartech Power, Inc.
SPECIFICATIONS

ELECTRICAL CHARACTERISITICS			
	SPM055P		SPM050P
Power at STC(Pm)	55W		50W
Maximum power voltage(Vpm)	18.18V		17.46V
Maximum power current(Ipm)	3.1A		2.98A
Open circuit voltage(Voc)	22.1V		21.89V
Short circuit current(Isc)	3.31A		3.23A
Tolerance	±5%		±5%
Module efficiency	13.2%		12.0%
Standard test conditions	Temperature 25℃, Irradiance 1000W/m², AM=1.5		
Maximum system voltage	1000V DC	Series fuse rating	10A
Temperature coefficient	Isc:0.05%/℃	Voc:-0.36%/℃	Pmax:-0.46%/℃

Specifications

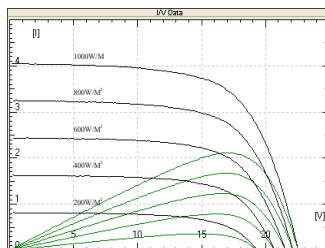
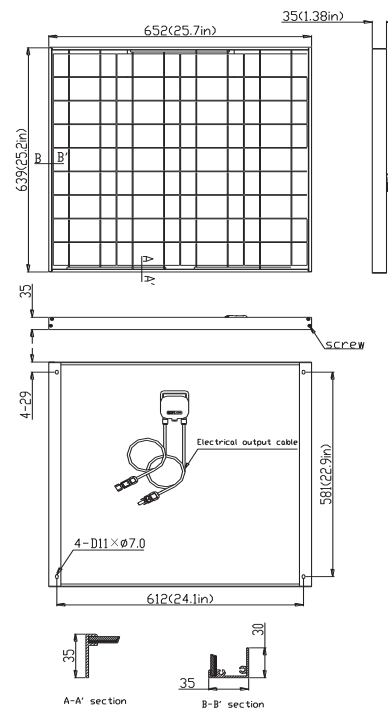
Cell	Multicrystalline silicon solar cell, (156mmx65mm)
No. of cell and connection	36(4x9)
Dimension	652(25.7in)mmx639(25.2in)mmx35(1.38in)mm
Weight	6.0KG

AMBIENT CONDITIONS

Operating Temperature	-40°C to +85°C
Storage Humidity	<90%

Output

Type of output terminal	Junction box
Cable	Length 900mm

IV CURVES**OUTLINE DIMENSIONS**

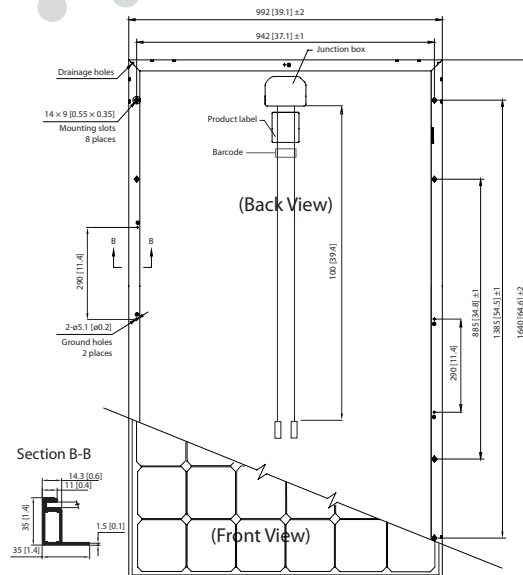
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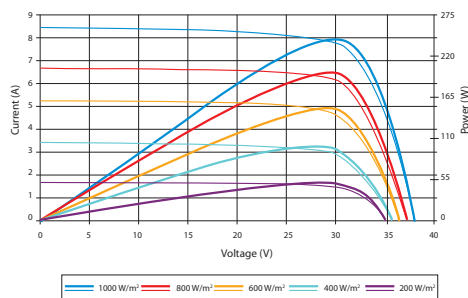
Solartech Power, Inc.
 1070 N. Kraemer Place
 Anaheim, CA 92806 USA
 Tel: 714-630-8880 Fax: 714-630-8887
www.solartechpower.com
 E-mail: sales@solartechpower.com

STP250S-20/Wd
STP245S-20/Wd
STP240S-20/Wd

SUNTECH

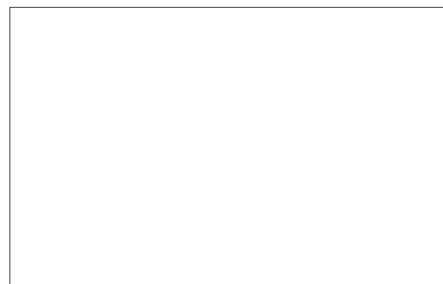


Current-Voltage & Power-Voltage Curve (240S-20)



Excellent performance under weak light conditions: at an irradiance intensity of 200 W/m² (AM 1.5, 25 °C), 95.5% or higher of the STC efficiency (1000 W/m²) is achieved

Dealer information



Electrical Characteristics

STC	STP250S-20/Wd	STP245S-20/Wd	STP240S-20/Wd
Maximum Power at STC (Pmax)	250 W	245 W	240 W
Optimum Operating Voltage (Vmp)	30.7 V	30.5 V	30.2 V
Optimum Operating Current (Imp)	8.15 A	8.04 A	7.95 A
Open Circuit Voltage (Voc)	37.4 V	37.3 V	37.2 V
Short Circuit Current (Isc)	8.63 A	8.52 A	8.43 A
Module Efficiency	15.4%	15.1%	14.8%
Operating Module Temperature	-40 °C to +85 °C		
Maximum System Voltage	1000 V DC (IEC)		
Maximum Series Fuse Rating	20 A		
Power Tolerance	0/+5 %		

STC: Irradiance 1000 W/m², module temperature 25 °C, AM=1.5;
 Best in Class AAA solar simulator (IEC 60904-9) used, power measurement uncertainty is within +/- 3%

NOCT	STP250S-20/Wd	STP245S-20/Wd	STP240S-20/Wd
Maximum Power at NOCT (Pmax)	183 W	180 W	177 W
Optimum Operating Voltage (Vmp)	27.9 V	27.8 V	27.7 V
Optimum Operating Current (Imp)	6.56 A	6.47 A	6.40 A
Open Circuit Voltage (Voc)	34.4 V	34.3 V	34.2 V
Short Circuit Current (Isc)	6.98 A	6.91 A	6.83 A

NOCT: Irradiance 800 W/m², ambient temperature 20 °C, AM=1.5, wind speed 1 m/s;
 Best in Class AAA solar simulator (IEC 60904-9) used, power measurement uncertainty is within +/- 3%

Temperature Characteristics

Nominal Operating Cell Temperature (NOCT)	45±2°C
Temperature Coefficient of Pmax	-0.44 %/°C
Temperature Coefficient of Voc	-0.34 %/°C
Temperature Coefficient of Isc	0.060 %/°C

Mechanical Characteristics

Solar Cell	Monocrystalline silicon 156 × 156 mm (6 inches)
No. of Cells	60 (6 × 10)
Dimensions	1640 × 992 × 35mm (64.6 × 39.1 × 1.4 inches)
Weight	18.2 kgs (40.1 lbs.)
Front Glass	3.2 mm (0.13 inches) tempered glass
Frame	Anodized aluminium alloy
Junction Box	IP67 rated (3 bypass diodes)
Output Cables	TUV (2Pfg1169:2007) 4.0 mm ² (0.006 inches ²), symmetrical lengths (-) 1000mm (39.4 inches) and (+) 1000 mm (39.4 inches)
Connectors	MC4 connectors

Packing Configuration

Container	20' GP	40' HC
Pieces per pallet	30	30
Pallets per container	6	28
Pieces per container	180	840

Information on how to install and operate this product is available in the installation instruction. All values indicated in this data sheet are subject to change without prior announcement. The specifications may vary slightly. All specifications are in accordance with standard EN 50380. Color differences of the modules relative to the figures as well as discolorations of/in the modules which do not impair their proper functioning are possible and do not constitute a deviation from the specification

Ultra-Light Marine Grade Solar Panels

45watt and 90watt Racing and Cruising Models

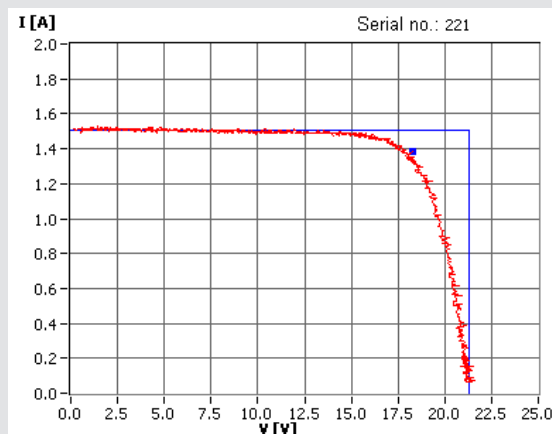
- **Racing models are the lightest panels on the market! Guaranteed.** The aluminum backplane provides excellent stiffness allowing us to make the ultimate light-weight panel. Highly efficient, for top performance when you need it most. Top sailors in the world agree!
- **Cruising models** are built with a sturdy aluminum frame. Perfect for a wide variety of mounting options. Same excellent output and performance as the racer model in a versatile package. Enjoy the safety of *glass free* panels on your boat!
- Custom panels are available allowing direct hull integration. Call us for further information.



*Panels specifications may change at any time.

Racing Model	45 watt	90 watt
Dimensions:	26 in x 21 in x .30 in 660mm x 533mm x 8mm	47.25in x 21in x .30in 1200mm x 533mm x 8mm
Weight:	4.5lbs / 2.0 kg	6.5 lbs / 2.94 kg
Rated Output:	45 watts	90 watts
Rated Voltage:	18 volts	18 volts
Price:	\$800 / 550€	\$1100 / 750€

Cruising Model	45 watt	90 watt
Dimensions:	26 in x 21in x 1.5in 660mm x 533mm x 38mm	47.25in x 21in x 1.5in 1200mm x 533mm x 38mm
Weight:	9 lbs / 4.0 kg	11 lbs / 4.98 kg
Rated Output:	45 watts	90 watts
Rated Voltage:	18 volts	18 volts
Price:	\$700 / 480€	\$1000 / 680€



www.IQSolar.com

15127 24th Avenue NE Suite 166 Redmond, WA 98052
206.334.2139

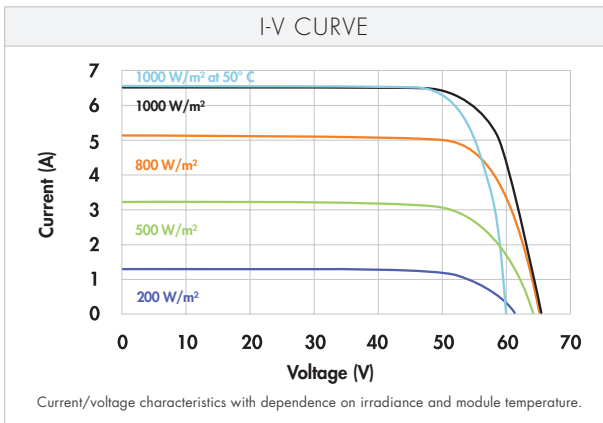
SUNPOWER™

E20/327 SOLAR PANEL

MODEL: SPR-327NE-WHT-D

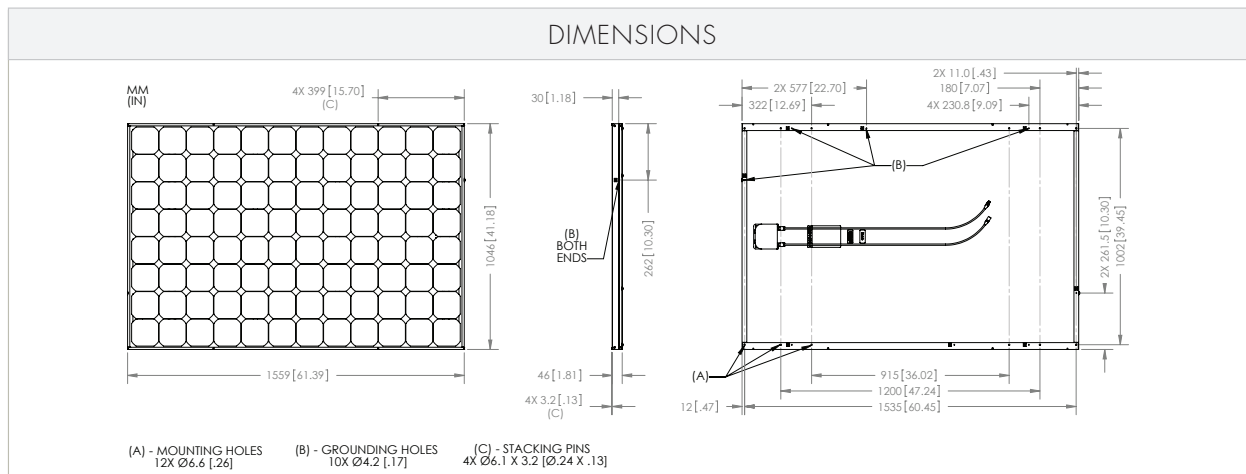
ELECTRICAL DATA		
Measured at Standard Test Conditions (STC): irradiance of 1000W/m ² , AM 1.5, and cell temperature 25° C		
Peak Power (+5/-3%)	P _{max}	327 W
Cell Efficiency	η	22.5 %
Panel Efficiency	η	20.1 %
Rated Voltage	V _{mpp}	54.7 V
Rated Current	I _{mpp}	5.98 A
Open-Circuit Voltage	V _{oc}	64.9 V
Short-Circuit Current	I _{sc}	6.46 A
Maximum System Voltage	UL	600 V
	IEC	1000 V
Temperature Coefficients	Power (P)	- 0.38 %/K
	Voltage (V _{oc})	-176.6 mV/K
	Current (I _{sc})	3.5 mA/K
NOCT		45° C +/- 2° C
Series Fuse Rating		20 A
Grounding		Positive grounding not required

MECHANICAL DATA	
Solar Cells	96 SunPower Maxeon™ cells
Front Glass	High-transmission tempered glass with anti-reflective (AR) coating
Junction Box	IP-65 rated with 3 bypass diodes Dimensions: 32 x 155 x 128 mm
Output Cables	1000 mm cables / MC4 compatible connectors
Frame	Anodized aluminum alloy type 6063 (silver); stacking pins
Weight	41.0 lbs (18.6 kg)



TESTED OPERATING CONDITIONS	
Temperature	- 40° F to +185° F (- 40° C to + 85° C)
Max load	113 psf 550 kg/m ² (5400 Pa), front (e.g. snow) w/specifed mounting configurations 50 psf 245 kg/m ² (2400 Pa) front and back (e.g. wind)
Impact Resistance	Hail: (25 mm) at 51 mph (23 m/s)

WARRANTIES AND CERTIFICATIONS	
Warranties	25 year linear power warranty 25 year limited product warranty
Certifications	Tested to UL 1703. Class C Fire Rating IEC 61215 Ed. 2, IEC 61730 (SCII)



Please read safety and installation instructions before using this product, visit sunpowercorp.com for more details.

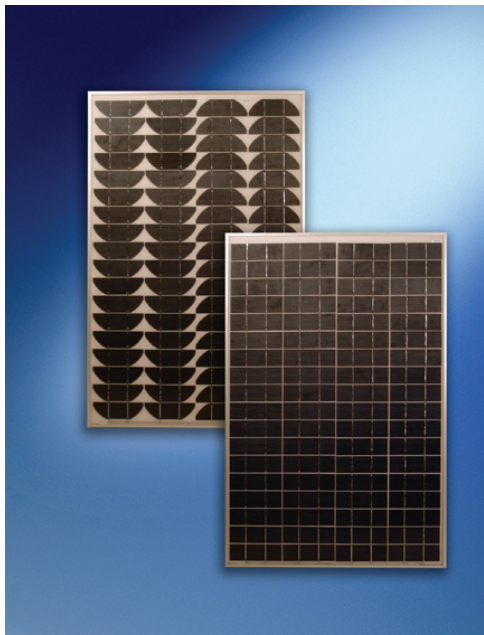
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Document #001-65350 Rev D / ITR_EN
CS 12_432

RELIABLE PEOPLE ::: RELIABLE PRODUCTS ::: RELIABLE POWER

SunWize® SW50A/55A/60A Solar Module



SunWize SW50A/55A/60A Series
shown with both solar cell configurations

SunWize SW50A/55A/60A solar modules deliver top-quality performance for all photovoltaic applications including rural electrification, water pumping, telemetry, communications, and general battery charging. SunWize modules can be used in single and multiple-module installations for charging nominal 12, 24 and 48 Volts DC battery banks. Each module consists of two strings of 36 solar cells connected in series providing maximum charging power.

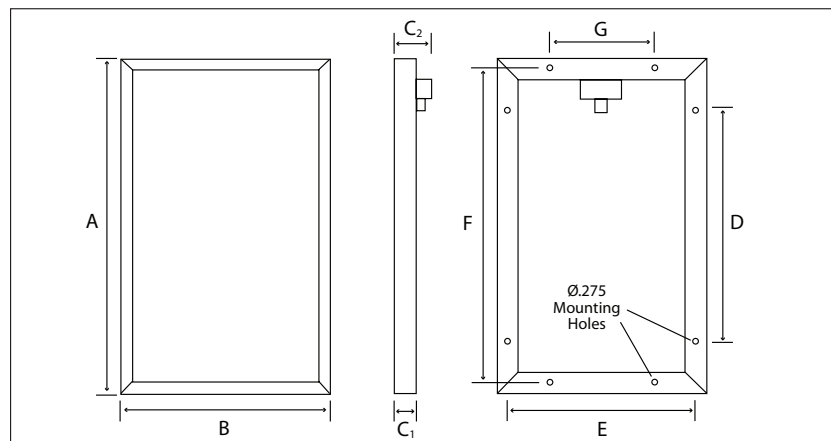
FEATURES:

- Single crystalline solar cells are encapsulated and bonded to the glass in multiple layers of ethylene vinyl acetate (EVA) and laminated with a white Tedlar™ backing insuring long life in severe environmental conditions.
- The glass surface is impact resistant and allows maximum light transmission.
- A weather resistant junction box accommodates all wiring methods including moisture-tight strain relief connectors and electrical conduit.
- Pre-drilled mounting holes.
- Bypass diodes within the junction box insure reliable operation.
- Anodized aluminum tubular frames add strength and durability to the modules.
- SW modules are manufactured in accordance to ISO 9002 standards, comply with U.S. and international standards, and are factory tested to verify superior performance. SW modules are ISPR certified to IEC 61215 and UL listed.
- All SunWize modules carry a 25-year, 80% power output warranty.

Model	Rated Power (Watts)	Rated Voltage (Vmp)	Rated Current (Imp)	Open Circuit Voltage (Voc)	Short Circuit Current (Isc)
SW50A	50	16.4V	3.05A	21.0V	3.40A
SW55A	55	16.7V	3.30A	21.0V	3.65A
SW60A	60	16.7V	3.60A	21.0V	3.95A

The operating temperature range is -40°C to +95°C in 100% relative humidity

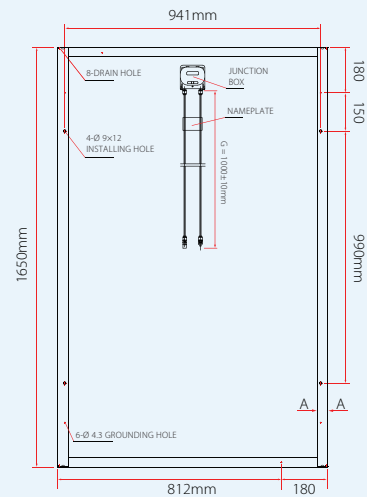
SunWize SW50A/55A/60A Solar Module		
MECHANICAL DIMENSIONS		
A	892 mm	35.11 inches
B	574 mm	22.59 inches
C₁	34 mm	1.34 inches
C₂	52 mm	2.05 inches
D	600 mm	23.62 inches
E	542 mm	21.34 inches
F	860 mm	33.86 inches
G	320 mm	12.60 inches
Weight:	6.0 kg.	13.22 lbs.


SunWize

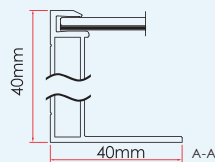
1155 Flatbush Road, Kingston NY USA 12401 Tel: 845-336-0146 • Fax: 845-336-0457 • www.sunwize.com

TSM-PC05.08 / TSM-PA05.08 THE UNIVERSAL SOLUTION

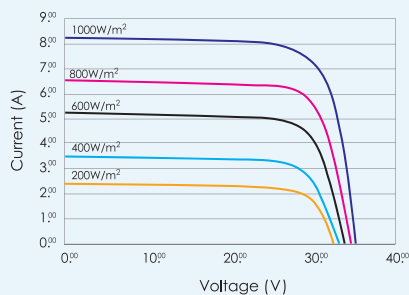
DIMENSIONS OF PV MODULE TSM-PC/PA 05.08



Back View



I-V CURVES OF PV MODULE TSM-230 PC/PA 05.08



Average efficiency reduction of 4.5% at 200W/m² according to EN 60904-1.

CERTIFICATION



ELECTRICAL DATA @ STC	TSM-225 PC/PA05.08	TSM-230 PC/PA05.08	TSM-235 PC/PA05.08	TSM-240 PC/PA05.08	TSM-245 PC/PA05.08
Peak Power Watts- P_{MAX} (Wp)	225	230	235	240	245
Power Output Tolerance- P_{MAX} (%)	0/+3	0/+3	0/+3	0/+3	0/+3
Maximum Power Voltage- V_{MP} (V)	29.4	29.8	30.1	30.4	30.7
Maximum Power Current- I_{MP} (A)	7.66	7.72	7.81	7.89	7.98
Open Circuit Voltage- V_{OC} (V)	36.9	37.0	37.1	37.2	37.3
Short Circuit Current- I_{SC} (A)	8.20	8.26	8.31	8.37	8.47
Module Efficiency η_m (%)	13.7	14.1	14.4	14.7	15.0

Values at Standard Test Conditions STC (Air Mass AM1.5, Irradiance 1000W/m², Cell Temperature 25°C).

ELECTRICAL DATA @ NOCT	TSM-225 PC/PA05.08	TSM-230 PC/PA05.08	TSM-235 PC/PA05.08	TSM-240 PC/PA05.08	TSM-245 PC/PA05.08
Maximum Power (W)	164	168	172	175	178
Maximum Power Voltage (V)	26.9	27.1	27.4	27.7	27.8
Maximum Power Current (A)	6.12	6.20	6.27	6.32	6.41
Open Circuit Voltage (V)	33.8	33.9	34.0	34.1	34.2
Short Circuit Current (A)	6.62	6.68	6.70	6.75	6.83

NOCT: Irradiance at 800W/m², Ambient Temperature 20°C, Wind Speed 1M/s.

MECHANICAL DATA

Solar cells	Multicrystalline 156 × 156mm (6 inches)
Cell orientation	60 cells (6 × 10)
Module dimension	1650 × 992 × 40mm (64.95 × 39.05 × 1.57 inches)
Weight	19.5kg (43.0 lb)
Glass	High transparency solar glass 3.2mm (0.13 inches)
Frame	Anodized aluminium alloy
J-Box	IP 65 rated
Cables / Connector	Photovoltaic Technology cable 4.0mm² (0.006 inches²), 1000mm (39.4 inches), MC4

TEMPERATURE RATINGS

Nominal Operating Cell Temperature (NOCT)	46°C (±2°C)
Temperature Coefficient of P_{MAX}	-0.43%/°C
Temperature Coefficient of V_{OC}	-0.32%/°C
Temperature Coefficient of I_{SC}	0.047%/°C

MAXIMUM RATINGS

Operational Temperature	-40~+85°C
Maximum System Voltage	1000V DC(IEC)/600V DC(UL)
Max Series Fuse Rating	15A

WARRANTY

10 year workmanship warranty
25 year linear performance warranty
(Please refer to product warranty for details)

PACKAGING CONFIGURATION

Modules per box: 24 pcs
Modules per 40' container: 672 pcs

TSM_EN_Feb_2012

Solar Laminate PVL-Series Model: PVL-144

UNI-SOLAR®

- High Temperature and Low Light Performance
- 5-Year Limited Product Warranty
- Limited Power Output Warranty:
92% at 10 years, 84% at 20 years, 80% at 25 years (of minimum power)
- Quick-Connect Terminals and Adhesive Backing
- Bypass Diodes for Shadow Tolerance

Performance Characteristics

Rated Power (P_{max}): 144 Wp
Production P_{max} Tolerance: $\pm 5\%$

Construction Characteristics

Dimensions: Length: 5486 mm (216"), Width: 394 mm (15.5"), Depth: 4 mm (0.2"),
16 mm (0.6") including potted terminal housing assembly
Weight: 7.7 kg (17.0 lbs)
Output Cables: 4 mm² (12 AWG) cable with weatherproof DC-rated quick-connect terminals
560 mm (22") length
Bypass Diodes: Connected across every solar cell
Encapsulation: Durable ETFE high light-transmissive polymer
Adhesive: Ethylene propylene copolymer adhesive sealant with microbial inhibitor
Cell Type: 22 triple junction amorphous silicon solar cells 356 mm x 239 mm
(14" x 9.4") connected in series

Qualifications and Safety

UL 1703 Listed by Underwriters Laboratories for electrical and fire safety (Class A Max. Slope 2/12, Class B Max. Slope 3/12, Class C Unlimited Slope fire ratings) for use in systems up to 600 VDC.

TÜV Rheinland IEC 61646 and IEC 61730 certified by TÜV Rheinland for use in systems up to 1000 VDC.

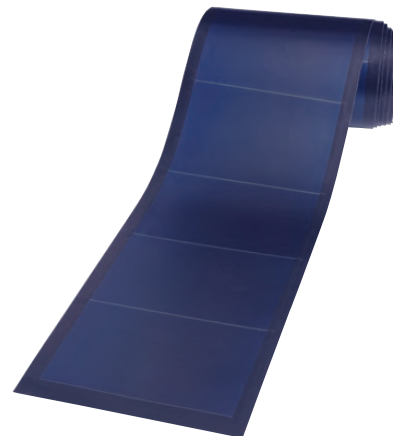
Laminate Standard Configuration

Photovoltaic laminate with potted terminal housing assembly with output cables and quick-connect terminals on top.

Application Criteria*

- Installation temperature between 10 °C - 40 °C (50 °F - 100 °F)
- Maximum roof temperature 85 °C (185 °F)
- Minimum slope: 3° (1/2:12)
- Maximum slope 60° (21:12)
- Approved substrates include certain membrane and metal roofing products. See United Solar for details.

*Detailed installation requirements are specified in United Solar installation manuals.



Flexible



Lightweight



Durable



No-Glass



Shadow Tolerant



More kWh



High Temp Performance



Low Light Performance

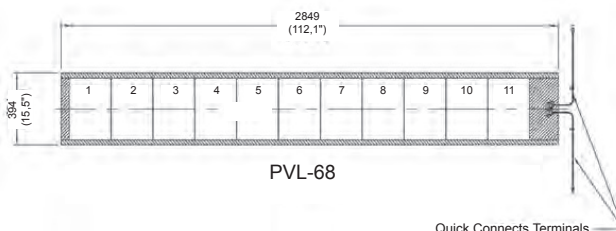
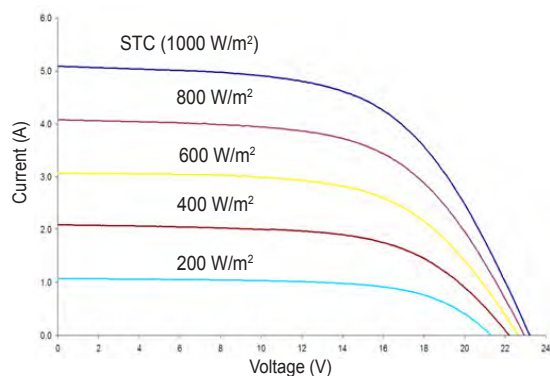
Technical Data Sheet

#AA5-3636-03

Solar Laminate PVL-Series Model: PVL-68

UNI-SOLAR®

IV Curves at various Levels of Irradiance at
Air Mass 1.5 and 25 °C Cell Temperature



All measurements in mm
Inches in parentheses
Tolerances: Length: ± 5 mm (1/4"), Width: ± 3 mm (1/8")

Electrical Specifications

STC

(Standard Test Conditions)
(1000 W/m², AM 1.5, 25 °C Cell Temperature)

Maximum Power (P_{max}): 68 W
Voltage at P_{max} (V_{mp}): 16.5 V
Current at P_{max} (I_{mp}): 4.13 A
Short-circuit Current (I_{sc}): 5.1 A
Open-circuit Voltage (V_{oc}): 23.1 V
Maximum Series Fuse Rating: 8 A

NOCT

(Nominal Operating Cell Temperature)
(800 W/m², AM 1.5, 1 m/sec. wind)

Maximum Power (P_{max}): 53 W
Voltage at P_{max} (V_{mp}): 15.4 V
Current at P_{max} (I_{mp}): 3.42 A
Short-circuit Current (I_{sc}): 4.1 A
Open-circuit Voltage (V_{oc}): 21.1 V
NOCT: 46 °C

Temperature Coefficients

(at AM 1.5, 1000 W/m² irradiance)

Temperature Coefficient (TC) of I_{sc} : 0.001/°K (0.10%/°C)
Temperature Coefficient (TC) of V_{oc} : -0.0038/°K (-0.38%/°C)
Temperature Coefficient (TC) of P_{max} : -0.0021/°K (-0.21%/°C)
Temperature Coefficient (TC) of I_{mp} : 0.001/°K (0.10%/°C)
Temperature Coefficient (TC) of V_{mp} : -0.0031/°K (-0.31%/°C)

$$y = y_{reference} \cdot [1 + TC \cdot (T - T_{reference})]$$

Notes:

- During the first 8-10 weeks of operation, electrical output exceeds specified ratings. Power output may be higher by 15%, operating voltage may be higher by 11% and operating current may be higher by 4%.
- Electrical specifications are based on measurements performed at standard test conditions of 1000 W/m² irradiance, Air Mass 1.5, and cell temperature of 25 °C after stabilization.
- Actual performance may vary up to 10% from rated power due to low temperature operation, spectral and other related effects. Maximum system open-circuit voltage not to exceed 600 VDC per UL, 1000 VDC per TÜV Rheinland.
- Specifications subject to change without notice.

Your UNI-SOLAR® Distributor:

Global Headquarters

United Solar Ovonic LLC
2956 Waterview Drive
Rochester Hills, MI 48309
Tel: 248.293.0440
Fax: 248.844.1214
Toll Free (USA): 1.800.528.0617
info@uni-solar.com

European Headquarters

United Solar Ovonic Europe SAS
Tour Albert 1er
65, avenue de Colmar
92507 Rueil-Malmaison Cedex
Tel: +33.1.74.70.46.24
Fax: +33.1.41.39.00.22
franceinfo@uni-solar.com

German Office

United Solar Ovonic Europe GmbH
Trakehner Strasse 7-9
60487 Frankfurt/Main
Tel: +49.69.7137667.0
Fax: +49.69.7137667.67
europeinfo@uni-solar.com

Italian Office

United Solar Ovonic Europe GmbH
Via Monte Baldo, 4
37069 Villafranca (VR)
Tel: +39.045.8600982
Fax: +39.045.8617738
italyinfo@uni-solar.com

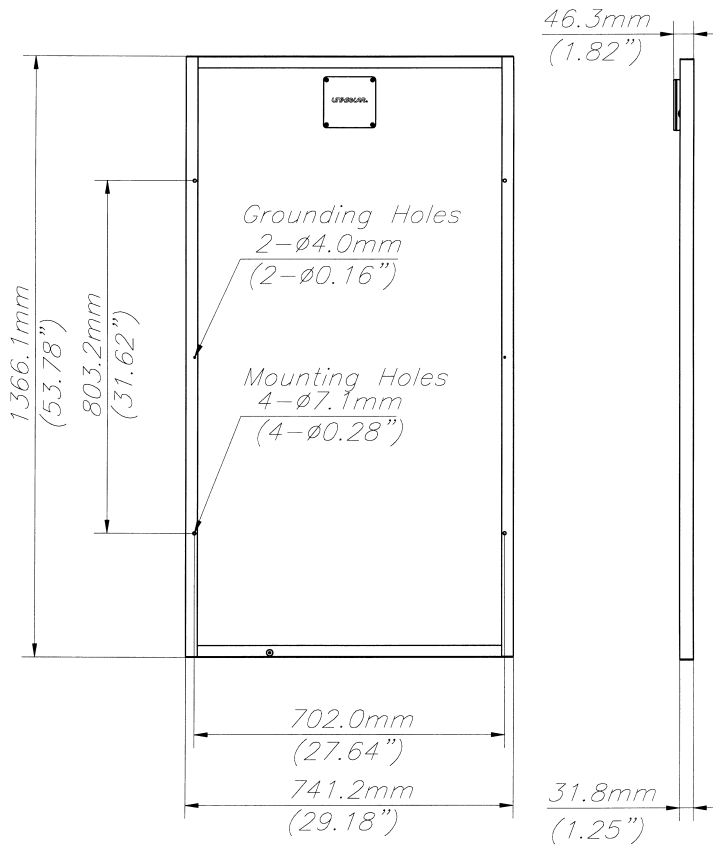
Spanish Office

United Solar Ovonic Europe GmbH
C/Jose Ortega y Gasset 25
28006 Madrid
Tel: +34 911269051
Fax: +34 911852670
spaininfo@uni-solar.com

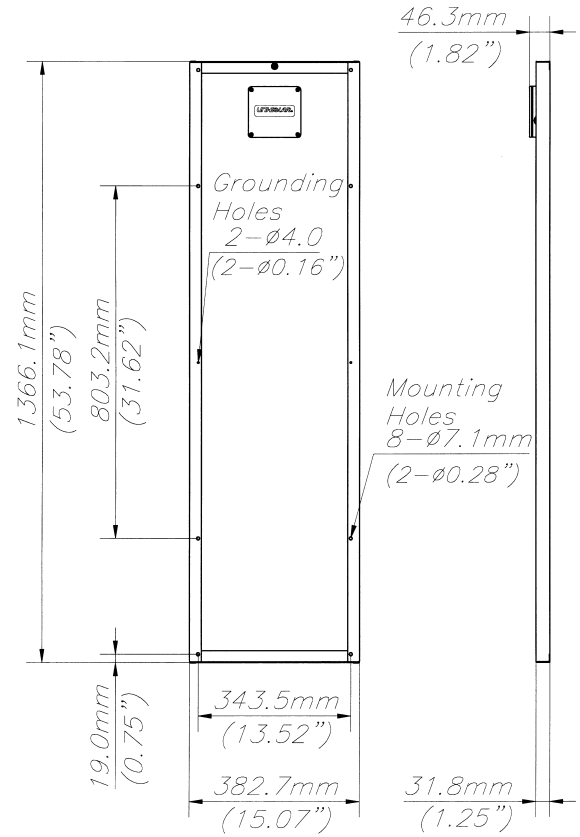
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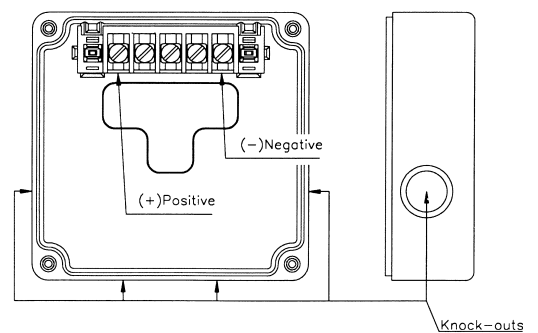
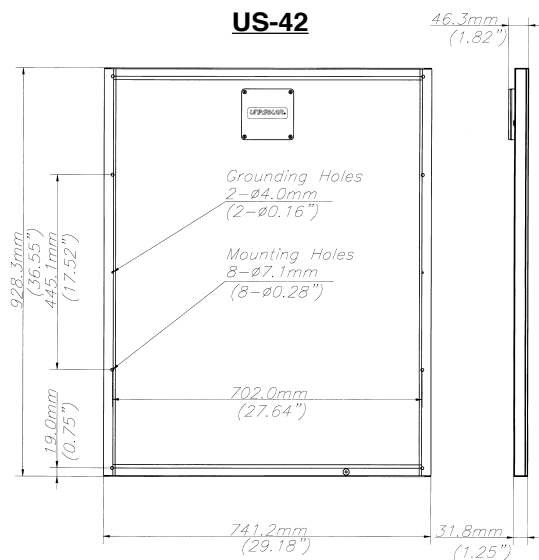
US-64



US-32



US-42



Specifications

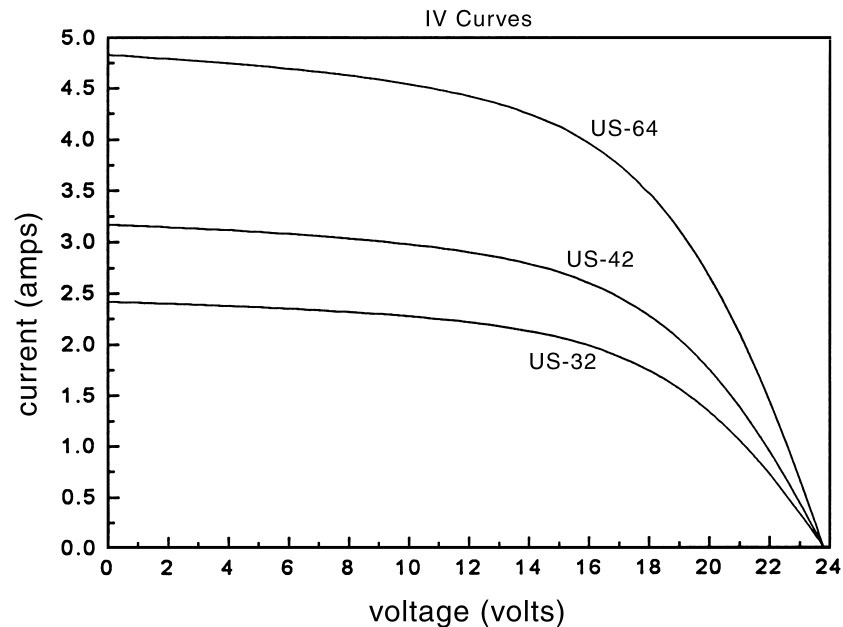
	US-64	US-42	US-32
Rated Power (Watts)	64	42	32
Operating Voltage (Volts)	16.5	16.5	16.5
Operating Current (Amps)	3.88	2.54	1.94
Open Circuit Voltage (Volts)	23.8	23.8	23.8
Open Circuit Voltage (Volts) at -10°C and 1250 W/m ²	27.1	27.1	27.1
Short Circuit Current (Amps)	4.80	3.17	2.40
Short Circuit Current (Amps)* at 75°C and 1250 W/m ²	6.30	4.20	3.10
Series fuse rating (Amps)	8.0	6.0	4.0
Minimum blocking diode (Amps)	8.0	6.0	4.0
Weight (lbs./kgs.)	20.2/9.17	13.8/6.27	10.6/4.8

During initial 8-10 weeks of operation, the module has higher electrical output than rated output. The output power may be higher by 15%, the operating voltage may be higher by 11% and operating current may be higher by 4%.

Electrical specifications ($\pm 10\%$) are based on measurements performed at standard test conditions of 1000 W/m² irradiance, Air Mass 1.5, and Cell Temperature of 25° C after long-term stabilization. Performance may vary up to 10% from rated power due to low temperature operation, spectral and related effects.

Maximum system open circuit voltage 600 VDC.

* Refer to section 690-8 of the National Electric Code for an additional factor of 125% which may be applicable.



Electrical Characteristics of US-64, US-42 and US-32 Modules at Standard Test Conditions of 1000 W/m² of AM 1.5 Irradiance and Cell Temperature of 25°C.

Specifications subject to change without notice.